

Lecture 6, Feb 02, 2026

6.5. Exponential Growth and Decay.

Goal. let $y(t)$ is the value of a quantity y at time t and if the rate of change of y with respect to t is proportional to its size $y(t)$ at any time, then

$$\frac{dy}{dt} = \kappa y \quad (I)$$

where κ is a constant. The equation (I) sometimes called the law of natural growth ($\kappa > 0$) or the law of natural decay ($\kappa < 0$).

Thm. The only solution of the differential equation $dy/dt = \kappa y$ are the exponential functions

$$y(t) = y(0) e^{\kappa t}$$

Pf. We know,

$$\frac{dy}{dt} = \kappa y$$

$$dy = \kappa y dt.$$

$$\frac{dy}{y} = \kappa dt.$$

$$\int \frac{1}{y} dy = \int \frac{dy}{y} = \int \kappa dt.$$

$$\frac{d}{dx}(\ln(x)) = \frac{1}{x}$$

$$\ln(y) = \kappa t + c.$$

$$e^{\ln(y)} = e^{\kappa t + c}.$$

$$y(t) = e^c \cdot e^{\kappa t}.$$

$$\text{let } t=0 \text{ then } y(0) = e^c \cdot e^{\kappa(0)} = e^c e^0 = e^c.$$

$$y(t) = y(0) \cdot e^{\kappa t}$$

* Radioactive Decay.

Radioactive substances decay by spontaneously emitting radiation. If $m(t)$ is the mass and m_0 the initial mass, then the model is given

$$\frac{dm}{dt} = km. \quad \text{where } k \text{ is a negative constant.}$$

$$m(t) = m_0 e^{kt}.$$

Example. The half-life of radium-226 is 1590 years.

- A sample of radium-226 has mass 100 mg. Find a formula for the mass of the sample that remains after t years.
- Find the mass remaining 1000 years correct to the nearest milligram.
- When will the mass be reduced to 30 mg?

Sol. We know that

$$m(t) = m_0 e^{kt}$$

We know that $m_0 = 100$ mg.

$$m(t) = 100 e^{kt}$$

In order to find k , we use the fact that $m(1590) = \underbrace{\frac{1}{2}(100)}_{m(t)} = 50$.

$$100 e^{1590k} = 50$$

$$e^{1590k} = \frac{50}{100} = \frac{1}{2}$$

$$\ln(e^{1590k}) = \ln\left(\frac{1}{2}\right)$$

$$1590k = -\ln(2)$$

$$k = \frac{-\ln(2)}{1590}$$

$$\begin{aligned} \ln\left(\frac{1}{2}\right) &= \ln(1) - \ln(2) \\ &= -\ln(2). \end{aligned}$$

then

$$m(t) = 100 e^{-\frac{\ln(2)t}{1590}}$$

b). The mass remaining after 1000 years is.

$$m(1000) = 100 e^{-\frac{\ln(2)(1000)}{1590}} \approx 65 \text{ mg.}$$

c). $30 = 100 e^{-\frac{\ln(2)t}{1590}}$ Solving for t .

$$\frac{30}{100} = e^{-\frac{\ln(2)t}{1590}}$$

$$\ln\left(\frac{3}{10}\right) = -\frac{\ln(2)t}{1590}$$

$$t = -\frac{1590 \ln\left(\frac{3}{10}\right)}{\ln(2)} \approx 2762 \text{ years.}$$

* Continuously Compounded interest.

Example. If \$5000 is invested at 2% interest, compounded annually then after 1 year the investment is worth $\$5000(1.02) = \5100.00 after 2 years it's worth

$$(\$5000 \times (1.02)) \times (1.02) = \$5000 (1.02)^2 = \$5202.00.$$

after t years it's worth.

$$\$5000 (1.02)^t = \$5000 \left(1 + \underbrace{0.02}_{2\%}\right)^t.$$

In general, if an amount A_0 is invested at an interest rate r then after t years it's worth is

$$A_0 (1+r)^t.$$

Usually, interest is compounded more frequently, say, n times per year

$$A = A_0 \left(1 + \frac{r}{n}\right)^{nt}.$$

For instance, after 3 years at 2% interest a \$5000 investment will be worth.

$$\$5000 (1.02)^3 = \$5306.04 \text{ with annual compounding.}$$

$$\$5000 \left(1 + \frac{0.02}{2}\right)^{2(3)} = \$5307.60 \text{ with semiannual compounding.}$$

$$\$5000 \left(1 + \frac{0.02}{4}\right)^{4(3)} = \$5308.39 \text{ with quarterly compounding}$$

$$\$5000 \left(1 + \frac{0.02}{12}\right)^{12(3)} = \$5308.92 \text{ with monthly compounding.}$$

You can see that the interest paid increase as the number of compounding periods (n) increases. If we let $n \rightarrow \infty$, then we will be compounding the interest continuously and the value of the investment will be

$$\begin{aligned} A(t) &= \lim_{n \rightarrow \infty} A_0 \left(1 + \frac{r}{n}\right)^{nt} \\ &= A_0 \lim_{n \rightarrow \infty} \left[\left(1 + \frac{r}{n}\right)^{n/r} \right]^{rt} \\ &= A_0 \left[\lim_{n \rightarrow \infty} \left(1 + \frac{r}{n}\right)^{n/r} \right]^{rt} \quad , \quad m = \frac{n}{r} \Rightarrow \frac{r}{n} = \frac{1}{m} \\ &= A_0 \left[\lim_{m \rightarrow \infty} \left(1 + \frac{1}{m}\right)^m \right]^{rt} \\ &= A_0 e^{rt} \end{aligned}$$

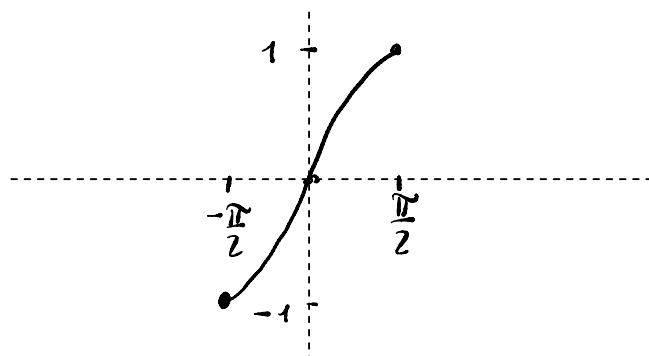
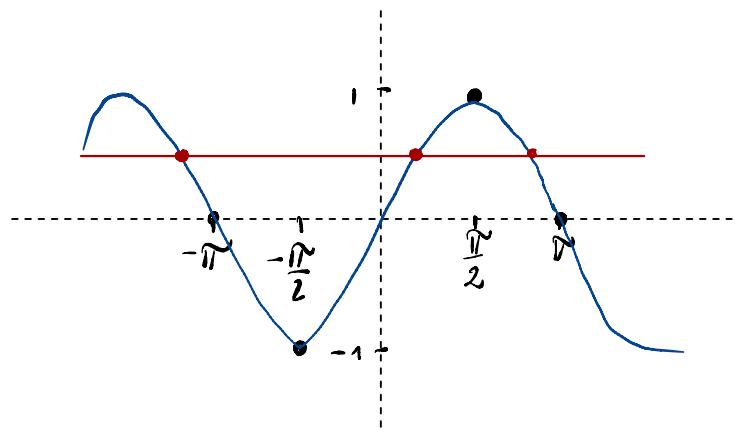
Returning to the example of \$5000 invested for 3 years at 2% interest, we see that with continuous compounding

$$A(3) = \$5000 e^{(0.02)3} = \$5309.18.$$

6.6 Inverse Trigonometric Functions.

let $f(x) = \sin x$.

let $y = \sin x$, $-\frac{\pi}{2} \leq x \leq \frac{\pi}{2}$.



Since the definition of an inverse function says that

$$f^{-1}(x) = y \stackrel{\text{def}}{\Leftrightarrow} f(y) = x. \quad \text{Notation:}$$

We have

$$\sin^{-1}(x) = \arcsin(x)$$

$$\sin^{-1}(x) = y \stackrel{\text{def}}{\Leftrightarrow} \sin y = x \quad \text{and} \quad -\frac{\pi}{2} \leq y \leq \frac{\pi}{2}.$$

Example. Evaluate $\sin^{-1}(\frac{1}{2})$, $\tan(\arcsin \frac{1}{3})$

Sln. • $\sin^{-1}(\frac{1}{2}) = \frac{\pi}{6}$ b/c $\sin(\frac{\pi}{6}) = \frac{1}{2}$ and $\frac{\pi}{6} \in [-\frac{\pi}{2}, \frac{\pi}{2}]$.

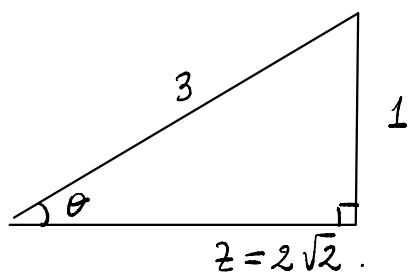
• let $\theta = \arcsin(\frac{1}{3})$ so, $\sin \theta = \frac{1}{3}$.

by the P-T

$$3^2 = 1^2 + z^2$$

$$9 - 1 = z^2 \Rightarrow z^2 = 8 \Rightarrow z = \sqrt{8} = 2\sqrt{2}$$

$$z = 2\sqrt{2}$$



We are looking for

$$\tan(\arcsin(\frac{1}{3})) = \tan(\theta) = \frac{1}{2\sqrt{2}}.$$

$$\sin^{-1}(\sin x) = x \quad \text{for} \quad -\frac{\pi}{2} \leq x \leq \frac{\pi}{2}$$

$$\sin(\sin^{-1}(y)) = y \quad \text{for} \quad -1 \leq y \leq 1.$$

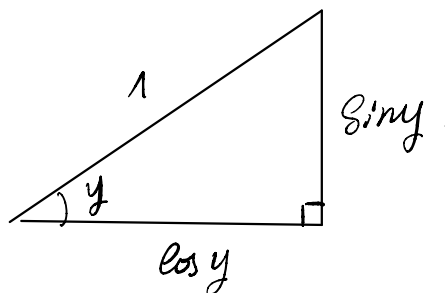
Let $y = \sin^{-1}(x)$ then $x = \sin y$. $-\frac{\pi}{2} \leq y \leq \frac{\pi}{2}$.

Differentiating $\sin y = x$

$$\frac{d}{dx}(\sin y) = \frac{d}{dx}(x)$$

$$\cos y \cdot \frac{dy}{dx} = 1$$

$$\frac{dy}{dx} = \frac{1}{\cos y}$$



by P-T

$$\sin^2 y + \cos^2 y = 1$$

$$\cos y = \sqrt{1 - \sin^2 y}$$

$$\cos y = \sqrt{1 - x^2}$$

$$\frac{d}{dx}(\sin^{-1}(x)) = \frac{1}{\sqrt{1-x^2}} \quad -1 < x < 1$$

Example. If $f(x) = \sin^{-1}(x^2 - 1)$. Find.

a). $\text{Dom } f$.

b). $f'(x)$.

c). $\text{Dom } f'$

Sln. a). Since the domain of the inverse sine function is $[-1, 1]$, the domain of f is

$$\begin{aligned} \{x \mid -1 \leq x^2 - 1 \leq 1\} &= \{x \mid 0 \leq x^2 \leq 2\} \\ &= \{x \mid 0 \leq |x| \leq \sqrt{2}\} \\ &= [-\sqrt{2}, \sqrt{2}] \end{aligned}$$

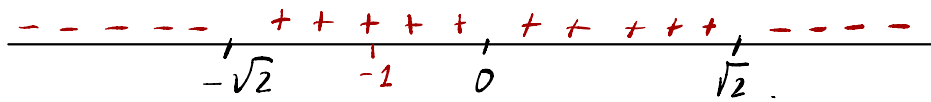
b). $f'(x) = \frac{1}{\sqrt{1 - (x^2 - 1)^2}} \cdot \frac{d}{dx}(x^2 - 1)$ by C-R.

$$= \frac{2x}{\sqrt{2x^2 - x^4}}$$

c). Solving $2x^2 - x^4 > 0$

$$x^2(2 - x^2) > 0$$

$$x^2(\sqrt{2} - x)(\sqrt{2} + x) > 0$$



$$\text{Dom } f' = (-\sqrt{2}, 0) \cup (0, \sqrt{2}).$$